Transcendence of a planar reflection–group growth series and its relaxed companion

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Abstract

Let $W = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the three sides of a right triangle in $\mathbb{Q}^2$ with positive unequal legs, and let $U(x) = \sum_{n \geq 0} u_n x^n$ be its geodesic word-length growth series — the integer sequence OEIS A396406 — with relaxed companion V. We prove:

(i) the catalytic building-block q-series $\Sigma_0, \Sigma_1, S_0, S_1$ are transcendental over $\mathbb{Q}(q)$, *unconditionally*: their integer coefficients grow super-polynomially along the squares, so $|q| = 1$ is a natural boundary by the Pólya–Carlson theorem;

(ii) the common growth rate is $\beta_2 = 1.4916177871\ldots$ *exactly*, and $\frac{3}{2}$ is excluded;

(iii) the relaxed series V is transcendental over $\mathbb{Q}(x)$; its one analytic input, a simple-saddle steepest-descent estimate, we prove here from first principles (Cauchy's theorem, Stirling, elementary Gaussian integration) rather than cite, with every constant explicit. For the true series U we prove: *conditional on a single amplitude estimate* ($\star$) *at the pole*, U is transcendental over $\mathbb{Q}(x)$, on the same analytic footing as V. By an exact machine-checked identity U reduces to V's estimate plus ($\star$); the amplitude has an elementary closed-form expansion with exact rational constants (Remark 7), and matching its zero against the travel-pole equation gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the gate threshold $1/\sqrt{2}$ — which *establishes* ($\star$) *to leading order* and at the first 70 poles; its residual $O(\tau)$ term we verify numerically but do not prove. The turning-point amplitude atom on the U-side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib* (Lemma 11);

(iv) an orthogonal, analysis-free route via Christol's theorem would settle U without any of the above: the p-kernel of ($u_n \bmod p$) is machine-verified to be maximally non-automatic at $p = 3, 5$, though a proof for dense-support series is open.

Thus V rests on self-contained analysis, and U on the same analysis plus the single residual ($\star$): every coefficient is derived elementarily and machine-verified, and the remainders are at the standard simple-saddle level — the precise rigor standing is stated in the text. In sharp contrast with the rational growth of hyperbolic groups and the algebraic growth series of Parry's wreath products, the geometric word metric places this virtually-lamplighter group outside every algebraic class.

1 Introduction

A right triangle $T \subset \mathbb{Q}^2$ with legs $a \neq b$ has three sides; the reflections across them generate a group $W = \langle R_0, R_1, R_2 \rangle \leq \mathrm{Aff}(\mathbb{R}^2)$. Its *growth series* $U(x) = \sum_n u_n x^n$ counts group elements by

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minimal word length in $\{R_0, R_1, R_2\}$. Because the legs are unequal the dihedral angles are irrational multiples of π (Niven), so W is a generic triangle group [7] rather than a Coxeter group, and — unlike the rational Coxeter case governed by Steinberg's formula [15] — its growth is not a priori rational. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \le n \le 9$ and first deviates at $n = 10$. This paper settles the arithmetic nature of both series: the relaxed companion V is transcendental over $\mathbb{Q}(x)$ with every analytic input proved self-contained, and the true series U is transcendental over $\mathbb{Q}(x)$ conditional on a single explicit amplitude estimate $(\star)$ at the pole — derived to leading order and verified at the first 70 poles, its $O(\tau)$ residual the sole remaining gap (§4; Remarks 6, 7, and 14).

Context. For word-growth series of groups, rationality is the norm in the classical families: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [3]), Coxeter systems are governed by Steinberg's rational formula, and for wreath products — including lamplighter-type groups — Parry computed the growth series for natural generating sets and found them *algebraic* [14]. Provably transcendental growth series are rare; the classical example is Stoll's [16]: the higher Heisenberg groups admit both rational and transcendental growth series, the transcendental one arising from a carefully constructed generating set. The group studied here is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$ [1], and its generating set is not constructed but imposed by Euclidean geometry — the three side-reflections of a right triangle. The results below show that for this natural geometric system the growth series is transcendental, by an explicit mechanism (a natural boundary for the building blocks; an accumulation of poles for the series themselves). This places the example in sharp contrast with the algebraicity of Parry's wreath-product series: the geometric word metric moves a virtually-lamplighter group outside every algebraic class. The higher-dimensional picture is subtler than a clean dimensional dichotomy [2]: the abstract right-angled Coxeter envelope of an n-orthoscheme has rational growth (rate $r_n = 1 + 2\cos\frac{2\pi}{n+3}$), but the geometric group is an amenable proper quotient that matches this rational envelope only to a finite depth before deviating below it. The plane is the one dimension in which the deviation has been computed past its onset — there it is the transcendental series treated here — and whether every higher dimension is likewise eventually transcendental is an open problem.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\ge 0}$ a cycle count. The relaxed counting function is the companion V; one has $v_n \ge u_n$, equality for $n \le 7$, first strict at $n = 8$. Both are governed by a linear q-difference (dilation $t \mapsto q^2 t$) *catalytic functional equation* (§2) that is q-holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q-series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2-gapped coefficients. Their coefficients grow super-polynomially, so Pólya–Carlson yields unconditional transcendence over $\mathbb{Q}(q)$ (§3). Both V and U acquire poles at the *travel poles* q_m, the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). That estimate is proved *self-contained* by a two-saddle steepest-descent argument — no asymptotic theorem is cited — and it is the *single* analytic input, shared by V (§5) and U (§6).

The U-side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q-Bessel function $J^{(3)}_{3/2}$, the transcendence gate reduces to one amplitude bound on that series at the travel poles. This bound is *derived* (Remark 7): the series collapses exactly to sinh-products, its expansion has exact rational constants, the pole equation collapses to the *same* sinh-family, and the phase-matching of the two conditions gives the pole–zero offset $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$ exactly — hence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the gate threshold. (The turning-point atom feeding this analysis, the rational envelope $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$ of Lemma 11, is formally verified in Lean 4.) The special-function obstacle that previously separated U from V is gone: the two series stand together, on the same self-contained analysis.

Section 7 records an independent arithmetic route: by Christol's theorem, if (u_n mod p) is not p-automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p-kernel is maximally non-automatic at $p = 3, 5$.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q-difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q,

$$\Sigma_1(q) = \sum_{i\geq 1}(-1)^{i-1}\frac{w^{2i}}{(2i)!}\,\rho_{i-1}, \quad S_e(q) = \sum_{j\geq 0}\frac{(-2(1-q))^j q^{j(j+1)}}{(q;q)_{2j}}, \tag{1}$$

and their companions Σ_0, S_0, where $\rho_j = q^j\,\eta_j$ with $\eta_j = e^{-B_j}$ a Gaussian-decaying form factor. Both V and U are rational functions of these blocks over $\mathbb{Q}(x)$; concretely the relaxed travel resolvent is $G_0^T = \Sigma_0/(1-\Sigma_1)$, and U inherits its denominator $1-\Sigma_1$ (Proposition 1). The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \qquad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$.

Proposition 1 (Invariance of the travel block)**.** *The denominator* $1 - \Sigma_1$ *is the identical, cycle-marker-free object in the generating functions of* U *and of* V*; hence* U *and* V *share the travel poles* q_m*.*

The proof is a strand-walk argument (every pure-travel element has cycle count 0, so the two length gradings agree on it); see the supplement.

3 Unconditional transcendence of the blocks

Theorem 2. *Each of* $\Sigma_0, \Sigma_1, S_0, S_1$ *is transcendental over* $\mathbb{Q}(q)$*.*

Proof. Each block is an integer power series with radius of convergence exactly 1. Along the squares its coefficients grow super-polynomially: for Σ_1 one has $|[q^{(j+1)^2}]\Sigma_1| \geq 2^{j+1}$, from the j^2-gapped catalytic telescoping (the block's coefficients $2, -4, 14, -52, 178, -856, 4626, \ldots$ at $q^{(j+1)^2}$, and the bound, are reproduced from the block's A/C recursion and machine-checked for $j \leq 9$ in Lean 4, `PolyaCarlson.lean`, theorem `coeff_bound`). A power series with integer coefficients and radius 1 is, by the Pólya–Carlson theorem [4], either a rational function whose poles are roots of unity or has the unit circle as a natural boundary. Super-polynomial coefficient growth rules out the rational alternative, so $|q| = 1$ is a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. □

This is the unconditional core: it uses no analytic gate. It does not, however, transfer directly to V or U, which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below.

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n\geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1}\zeta(2n)}{n(2\pi)^{2n}}$, and set $B_s = \sum_{n\geq 1} \varphi_n \tau^{2n}(P_n(2s) - s)$, where $P_n(M) = \sum_{m=1}^{M} m^{2n}$ is the Faulhaber sum. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = we^{-\tau/2}$ and

$$T_2 = \sum_{n\geq 1} (-1)^n \, g_n \, \frac{W^{2n}}{\Gamma(2n+1)}, \qquad g_s = 1 - e^{-B_s}. \tag{2}$$

Everything hinges on $T_2 \to 0$, and on its sharp size $|T_2| = O(\sqrt{\tau})$.

Lemma 3 (amplitude regularity). *On the strip $\mathcal{S} = \{0 \leq \Im s \leq W/2,\ \mathrm{Re}\, s \geq 0\}$ one has, for $\tau < \pi^2/8$,*

$$\mathrm{Re}\, B_s \ \geq\ -\tfrac{\sqrt{2}}{18}\sqrt{\tau} - 0.8\,\tau^{3/2},$$

so $A := 1 - e^{-B_s}$ is bounded ($|A| \leq 2.05$) and of bounded variation on $\mathcal{S}$.

Proof sketch. Since $W/2 < \pi/\tau$ there is no resonance: writing B_s through the Γ-function form, the divergent $-\pi^2 k/\tau$ Stirling terms cancel exactly between the s-shifted and boundary pairs, so B_s is finite on $\mathcal{S}$. The lower bound is set by the $n = 1$ term $\frac{1}{36}\tau^2\, \mathrm{Re}(4s^3 + 3s^2 - s)$, which minimises on $\mathcal{S}$ at the diagonal $s = \frac{W}{2}(1+i)$ to $-\frac{\sqrt{2}}{18}\sqrt{\tau} + \frac{5\sqrt{2}}{72}\tau^{3/2} \geq -\frac{\sqrt{2}}{18}\sqrt{\tau}$ (the factor $e^{-3\tau/2}$ in W makes the subleading correction *positive*). The tail is controlled by an explicit geometric majorant: for $n \geq 2$, $|\varphi_n| \leq \zeta(4)/(n(2\pi)^{2n})$ and $|P_n(2s)| \leq 2|2s|^{2n+1}/(2n+1)$ (the Bernoulli-polynomial bound $|B_{2n+1}(z)| \leq 2|z|^{2n+1}$, $|z| \geq 2n+1$), so on $|s| \leq 2W$

$$\Big|\sum_{n\geq 2} \varphi_n \tau^{2n}(P_n(2s) - s)\Big| \leq \sum_{n\geq 2} \frac{2\zeta(4)}{n(2n+1)} |2s| \Big(\tfrac{|s|\tau}{\pi}\Big)^{2n} \leq 0.8\,\tau^{3/2},$$

a convergent geometric series in $(|s|\tau/\pi)^2 \leq 8\tau/\pi^2 < 1$; for $|s| > 2W$ the real-axis growth of $\log\Gamma(2s+1)$ forces $\mathrm{Re}\, B_s > 0$. Bounded variation follows because differentiating the majorant term-by-term multiplies the n-th term by $O(n)$ and preserves convergence. Full constants are verified in the supplement (`lemBbounded_tail.py`). □

Rather than cite the general steepest-descent theorem, we prove the estimate directly; the argument exhibits the saddle explicitly and uses only Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3 (it reproves, for this integrand, the specialization of [13, Ch. 4, Thm. 7.1]).

Lemma 4 (decay of T_2). *As $q \to 1^-$, $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\, \sin w + O(\tau)$; in particular $|T_2| = O(\sqrt{\tau}) \to 0$.*

Self-contained proof by steepest descent. Since $\cos W = \sum_{n\geq 0}(-1)^n W^{2n}/(2n)!$, (2) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n\geq 0}(-1)^n e^{-B_n} W^{2n}/(2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z}/\Gamma(2z+1)$, analytic for $\mathrm{Re}\, z > -\frac{1}{2}$ (each term of B_z is a polynomial in z). As $\pi \csc \pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i}\oint b(z)\, \pi \csc(\pi z)\, dz$ over a contour enclosing $\{0, 1, 2, \dots\}$. Write the integrand as $e^{h(z)}\pi \csc \pi z$ with $h(z) = 2z \log W - \log \Gamma(2z+1)$; including the csc phase, the exponent is stationary where $2 \log W - 2\psi(2z+1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_\pm^* = \pm \tfrac{i}{2} W\big(1 + O(1/W)\big), \qquad h''(z_\pm^*) = \pm\tfrac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through $z^*_\pm$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \to +\infty$ and csc suppresses $|\Im z| \to \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly* — the divergent factors $e^{\pm\pi W/2}$ from $\csc(\pi z^*_\pm)$ and from $1/\Gamma(\pm iW+1)$ cancel identically, each saddle contributing $\frac{1}{2}e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z^*_\pm}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$$B_{z^*_+} = \varphi_1\tau^2\big(P_1(iW) - \tfrac{iW}{2}\big) + O(\tau^{3/2}) = \tfrac{\tau^2}{24}\Big(-\tfrac{i}{3}W^3\Big) + O(\tau^{3/2}) = -i\tfrac{\sqrt{2}}{36}\sqrt{\tau} + O(\tau)$$

(using $\tau^2 W^3 = 2\sqrt{2}\sqrt{\tau}$), and $B_{z^*_-} = \overline{B_{z^*_+}}$. Hence

$$S_e = \tfrac{1}{2}e^{-B_{z^*_+}}e^{iW} + \tfrac{1}{2}e^{-B_{z^*_-}}e^{-iW} + E = \cos W - \tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + E.$$

The remainder E is $O(\tau)$: parametrising the descent path by arclength s, the integrand equals $e^{h(z^*)}e^{-s^2}\big(1+O(sW^{-1/2})+O(\tau)\big)$, and term-by-term Gaussian integration (the odd powers vanishing) leaves an error controlled by the next saddle coefficient $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, the total variation of e^{-B_z} along the path being bounded by Lemma 3. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. □

Remark 5 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2\tau}$ and linearising the Gaussian through $e^{-j^2\tau} = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)}e^{iju}\,du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}}\int_{-\infty}^{\infty} e^{-u^2/(4\tau)}\,\Psi(u,q)\,du, \qquad \Psi(u,q) = \sum_{j\ge 0}\frac{\big(-2(1-q)q\,e^{iu}\big)^j}{(q;q)_{2j}}, \tag{3}$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j/(q;q)_{2j} < \infty$, $Z = -2(1-q)q\,e^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u,q) = \overline{\Psi(u,q)}$, so (3) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)}\operatorname{Re}\Psi\,du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)}e^{iju}\,du = \sqrt{4\pi\tau}\,e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`).

We caution that this representation, though exact and convenient, is *not* an analytic shortcut — a point where an earlier draft overreached. The amplitude $\Psi(\cdot,q)$ is entire in ζ, but it is *not* uniformly bounded as $q \to 1^-$: on any fixed window it grows like $\cosh\big(w\sin(u/2)\big)$ (numerically $|\Psi|$ already exceeds 300 at $u = 1,\ \tau = 10^{-2}$), and its termwise limit $\cos(w\,e^{iu/2})$ *differs* from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q;q)_{2j}$ is itself $O(1)$; numerically $\Psi/\cos(we^{iu/2}) \approx 0.62$). Estimating S_e is therefore a genuine competition between this exponential growth and the Gaussian — a simple *complex* saddle at $u^* \approx -\sqrt{2\tau}$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses. In particular the bounded-variation input of Lemma 3 is *not* eliminated by passing to (3); it, or the equivalent simple-saddle estimate of Lemma 4, remains the analytic core. The value of (3) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in the supplement (`gaussint_verify.py`).

Remark 6 (an elementary derivation of the coefficients of Lemma 4). The leading term and the constant $\sqrt{2}/36$ of Lemma 4 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1-e^{-k\tau}=2e^{-k\tau/2}\sinh(k\tau/2)$ throughout, the j-th term of $S_e=\sum_{j\ge 0}(-2(1-q))^j q^{j(j+1)}/(q;q)_{2j}$ collapses to

$$S_e=\sum_{j\ge 0}(-1)^j e^{-j\tau}\,\frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j}\sinh(k\tau/2)}=\sum_{j\ge 0}(-1)^j\,\frac{(2/\tau)^j}{(2j)!}\,e^{-j\tau+E_j},\quad E_j=\tfrac{j\tau^2}{24}-\tfrac{\tau^2}{144}(2j)(2j+1)(4j+1)+O(\tau^4 j^5),$$

E_j being the exact sinh-curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j(-1)^j(2e^{-\tau}/\tau)^j/(2j)!=\cos W$ with $W=\sqrt{2/\tau}\,e^{-\tau/2}$, which is exactly the frequency of Lemma 4 — so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j(-1)^j y^j j^m/(2j)!=(y\,d/dy)^m\cos\sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9}(y\,d/dy)^3\cos\sqrt{y}=-\frac{\tau^2}{9}\cdot\frac{W^3}{8}\sin W=-\frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W$ (using $\tau^2W^3=2\sqrt{2}\,\sqrt{\tau}$), giving

$$S_e=\cos W-\tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W+O(\tau),\qquad W=\sqrt{2/\tau}\,e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k\sin W$, $W^k\cos W$, so it serves the $O(\tau^2)$-relative demand of the U-numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate — that the alternating expansion is genuinely asymptotic with $O(\tau)$ error. That step is irreducibly a *cancellation* statement, and demonstrably so: not only does the triangle inequality on $\sum_j(2/\tau)^j/(2j)!\,|\cdot|$ return $\cosh W\asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J|S_J|=9,\ 48,\ 1459,\ 7.6\times10^4$ at $\tau=0.1,0.05,0.02,0.01$). Hence *every* partial-summation method — Abel, summation-by-parts, Euler transformation — fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Lemma 4: the saddles $z^*_\pm=\pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument render Lemma 4 independent of any cited asymptotic theorem, resting only on Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3.

Remark 7 (the same machinery derives the U-numerator amplitude). The single U-side input — the bound $|\sum_k d_k|\le\frac{3}{8\sqrt{2}}\tau^{5/2}$ at the travel poles on the one explicit series $P_{12}=\frac{2q^3}{1-q^3}\sum_k d_k$, $d_k=(-2)^k(1-q)^k q^{k^2+3k}/[(q^2;q^2)_k(q^5;q^2)_k]$ — submits to the identical reduction. Writing $1-q^m=2e^{-m\tau/2}\sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k=(-1)^k e^{-k\tau}\,\frac{\sinh^{k+1}(\frac{\tau}{2})\,\sinh(\frac{3\tau}{2})\,\sinh((k{+}1)\tau)}{\prod_{m=1}^{2k+3}\sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities $k^m\mapsto(y\,d/dy)^m$) yields, with $\Phi(y)=(\sin W-W\cos W)/(2W^3)$, $W=\sqrt{y}$, $D=y\,d/dy$,

$$\sum_k d_k=6\Big[\Phi+c_3D^3\Phi+c_2D^2\Phi+\tfrac{c_3^2}{2}D^6\Phi+(c_5+c_2c_3)D^5\Phi+\tfrac{c_3^3}{6}D^9\Phi\Big](y_*)+O(\tau^3),$$

$y_*=\frac{2}{\tau}e^{-\tau-23\tau^2/36}$, $c_2=-\frac{5}{12}\tau^2$, $c_3=-\frac{1}{9}\tau^2$, $c_5=\frac{1}{450}\tau^4$ — *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau\in[0.005,0.08]$ and the formula reproduces the exact series at the poles $m = 6, \dots, 32$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \to 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the gate constant is $(3\tau/2)\,\delta w/\tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1 - \Sigma_1 = \sum_{j\ge 0}(-1)^j \sinh^j(\frac{\tau}{2})/\prod_{m=1}^{2j}\sinh(\frac{m\tau}{2})$ — the S_e-series stripped of its $e^{-j\tau}$ phase — so the identical expansion applies with $y_* = \frac{2}{\tau}e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M + \frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} + \frac{41\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2), \qquad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} - \frac{259\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically* — both are produced by the universal cubic curvature c_3, and the constant $\sqrt{2}/36$ is that of Lemma 4; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8}\,\tau^{3/2} + O(\tau^2), \qquad \text{whence} \qquad |\sum_k d_k| = \frac{3\tau}{2}\,\delta w\,(1 + O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}}\tau^{5/2}(1 + O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin.* Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 10, 16, 22, 28$, and $\delta w/\tau^{3/2} = 0.17699,\ 0.17687,\ 0.17683,\ 0.17681 \to \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Lemma 4: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

5 The relaxed series V

Theorem 8. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$.*

Proof. V contains the travel resolvent $(1 - \Sigma_1)^{-1}$ times a bulk-dressing factor holomorphic and non-vanishing at the q_m; thus V has a genuine pole at each travel pole q_m, and the $q_m \uparrow 1$ accumulate at the boundary of the disc of convergence. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$; an infinite pole set accumulating at the boundary is impossible for an algebraic function, so V is transcendental — *provided* the poles are not cancelled by the numerator. Non-cancellation is exactly the statement that $S_e(q_m) = \cos W - T_2 \neq 0$ for all large m, i.e. that $|T_2| < |\cos W|$ there; since $|\cos W| \asymp \sqrt{\tau/2}$ at the poles while $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$ (Lemma 4), the inequality holds with a fixed margin. Hence V is transcendental; its sole analytic input is the simple-saddle estimate of Lemma 4, which we prove here from first principles (Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3), rather than by appeal to a general theorem. □

Corollary 9. *The exponential growth rate of both u_n and v_n is $\beta_2 = 1.4916177871\ldots$, the reciprocal of the smallest travel pole; the value $\frac{3}{2}$ is excluded. This is unconditional (it needs only the location of the dominant travel pole, not Lemma 4).*

Remark 10 (the number β_2: a value-transcendence question). Whether the *number* β_2 is transcendental is open, but the question has a clean reduction. Since $\beta_2 = 1/\sqrt{q^*}$ with $\Sigma_1^T(q^*) = 1$, the number β_2 is algebraic iff q^* is; if q^* were algebraic, the series Σ_1^T would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko's theorem on Eisenstein values [11], $\sum_n \alpha^{n^2}$ is transcendental at *every* algebraic α with $0 < |\alpha| < 1$ [8]. The blocks here are theta-*like* — their coefficients ride the squares $q^{(j+1)^2}$, which is the source of the natural boundary of Theorem 2 — but their Euler-product denominators $(q;q)_{2k+1}$ spread the support densely, and no analogue of [11] is known off the modular class. Thus β_2 transcendental would follow from a value-transcendence theorem for this partial-theta class at algebraic points; we record this as the precise open question. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

6 The true series U

U inherits the travel poles of Proposition 1; it is transcendental once its numerator is non-zero at infinitely many q_m. The Bousquet-Mélou–type transfer expresses the relevant residue through a Möbius function of a cocycle variable, and non-vanishing reduces to a *gate*

$$b_0 > 0 \qquad \text{and} \qquad s := \tfrac{q}{1-q}\, t_1 < 1, \qquad t_1 = \frac{P_{12}}{S_e}, \tag{4}$$

on the off-diagonal cocycle entry P_{12} (see the supplement, § lifting). Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1-q)q^{2n+2})y_n - q^3 y_{n-1}$; setting $x = q^n$ turns it into the linear q-difference equation

$$Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1-q)q^2x^2)Y(x), \tag{5}$$

whose regular solution is, in closed form, a Hahn–Exton q-Bessel function,

$$Y_3(x) = \sum_{k\geq 0} d_k x^{2k+3}, \qquad d_k = \frac{(-2)^k(1-q)^k q^{k^2+3k}}{(q^2;q^2)_k(q^5;q^2)_k} = J^{(3)}_{3/2} \text{ normalised}, \tag{6}$$

and $P_{12} = \frac{2q^3}{1-q^3} Y_3(1)$. The denominator S_e is the block of (1), controlled by Lemma 4. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

6.1 The amplitude atom, formally verified

As $q \to 1$, equation (5) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2}J_{3/2}(Wx)$. The q-corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire* — it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 11 (two-Bessel confluence; amplitude normalisation)**.** *The regular solution of* (5) *has the confluent form*

$$Y_3(x) = \gamma_{\rm cl}\, x^{3/2}\Big[J_{3/2}(X) + \tfrac{\tau X}{2} J_{5/2}(X)\Big] + O(\tau^2), \qquad X = wx,\ \gamma_{\rm cl} = \tfrac{3\sqrt{\pi}}{4}\big(\tfrac{w}{2}\big)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\rm cl} = 1 + \tau\, h(X) + O(\tau^2)$ with the explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \ \in\ [1, \tfrac{3}{2}] \qquad (X > 0).$$

Consequently $\gamma/\gamma_{\rm cl} = 1 + O(\tau)$ with explicit constant $\frac{3}{2}$.

Proof. From the q-Pochhammer confluence $d_k = \tau^{-k} D_k(1-k\tau+O(\tau^2))$ with $D_k = (-1)^k/[2^k k!\,(\frac{5}{2})_k]$, the leading sum is $\gamma_{\rm cl} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2}\gamma_{\rm cl} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(;\frac{7}{2};\cdot)$. Both series are entire in X. The envelope of $J_{3/2} + \frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X(J_{3/2}J_{5/2} + Y_{3/2}Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X)J_{5/2}(X) + Y_{3/2}(X)Y_{5/2}(X) = \frac{4}{\pi X^2}\Big(1 + \frac{3}{2X^2}\Big),$$

which with $M_{3/2}^2 = \frac{2}{\pi X}(1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \le h \le \frac{3}{2}$. □

Remark 12 (machine verification). The cross-term identity and the bound $1 \le h(X) \le \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Theorem 13. *Suppose the numerator amplitude estimate* ($\star$) *holds: at every travel pole q_m the amplitude $Y_3(1/q_m)$ agrees with its confluent leading part to relative order $O(\tau^2)$ — equivalently $|P_{12}(q_m)| = O(\tau^{3/2})$. Then the true growth series U is transcendental over $\mathbb{Q}(x)$. Every* other *analytic input coincides with Theorem 8's: the reduction to the numerator gate* (4) *is exact and machine-checked, and the denominator half of the gate is Lemma 4, shared with V. The estimate* ($\star$) *is* derived to leading order *in Remark 7 — phase-matching the amplitude's zero against the travel-pole equation gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, whence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the gate threshold $1/\sqrt{2}$ — and it holds at the first* 70 *poles; its residual $O(\tau)$ term (the amplitude drift from turning point to pole) is established numerically but not proved, and is the sole gap (Remark 14).*

Proof. At a travel pole write $t_1 = (E/E_S)(1+R/E)/(1+R_S/E_S)$ where $E = \frac{1}{2}(w-W)^2 \sin w\, \sin(w-W)$ is the elementary confluent leading part of P_{12}, $E_S = \sin w \sin(w-W)$ that of S_e, and $R = P_{12} - E$, $R_S = S_e - E_S$ the residuals. The leading ratio $E/E_S = \frac{1}{2}(w-W)^2 \to \frac{\tau}{4}$ is exactly elementary, so $s \to \frac{1}{4} < 1$. *Denominator.* $R_S = \cos w \cos(w-W) - T_2$, and by Lemma 4 $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$, so $S_e = E_S + R_S$ stays bounded away from zero (the same margin as in Theorem 8) and R_S/E_S is bounded — exactly as for V. *Numerator.* The remaining bound $R/E = O(\tau)$ is equivalent to $|P_{12}| = O(\tau^{3/2})$ at the pole; since $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e$ is an $O(\sqrt{\tau})$ *cancellation*, it demands $Y_3(1/q)$ at $X = w$ to relative order $O(\tau^2)$. This bound holds at every computed pole ($|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.1768$, a $4\times$ margin below $1/\sqrt{2}$, over $m = 4, \ldots, 70$), but is established there *numerically*; it is exactly the hypothesis ($\star$). Under ($\star$), $t_1/\tau = \frac{1}{4} + O(\tau)$ and $s < 1$ with positive margin at every pole, so $\mathcal{N}_U(q_m) \neq 0$, and U inherits infinitely many uncancelled poles accumulating at $q = 1$, hence is transcendental. □

Remark 14 (the one remaining gap, stated honestly). Lemma 11 discharges the amplitude *through the turning point* $X = \frac{3}{2}$ — the obstruction to any Liouville–Green treatment, and the one piece formerly thought to settle the U-side. It does *not*, however, supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$-relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$ — i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$-relative — and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) is the genuine unknown. Because $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself; only $Y_3(1/q)$ to $O(\tau^2)$-relative — a uniform pole-amplitude resummation of the *same analytic class* as Lemma 4 — closes it. Crucially, this residual is of *exactly the denominator's kind*, not a harder one. The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q-Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N\, S_e - 1}{x_N} \qquad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e: one analytic class — that of Lemma 4 — not two. In this precise sense U sits on V's footing. What U additionally requires is the pole asymptotics of x_N, X_N — again Hahn–Exton, again governed by the exact q-Wronskian of Olde Daalhuis [6], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \to 1$ expansion is explicit via Moak's q-Stirling formula (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 7: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude's zero — both collapsing to the same sinh-product family — derives the gate value $1/(4\sqrt{2})$. U's analytic inputs thereby coincide with V's up to the $O(\tau)$ residual of ($\star$). (The orthogonal arithmetic route of §7 would settle U without any of this analysis, but is itself open.)

7 An orthogonal arithmetic route

For each prime p, $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p-automatic (Christol [5]), iff its p-kernel is finite. We find the p-kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p-ary tree with *no* deficit. (An apparent deficit of 1 at level 3 — $1, 4, 13, 39$ vs. the full $1, 4, 13, 40$ at $p = 3$ — is a *comparison-length artifact*: a level-k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-verified in Lean 4. For the true series itself, the 3-kernel of ($u_n \bmod 3$) through level 3 is the full ternary tree, deficit 0: all $1 + 3 + 9 + 27 = 40$ decimation words of length ≤ 3 are pairwise distinct — at comparison lengths 9 and 10 alike — on the 271-term prefix $n \leq 270$ (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms and byte-identical to the earlier validated $n \leq 180$ run on the overlap). For the sibling block Σ_1, already transcendental over $\mathbb{Q}(q)$ by Theorem 2, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the A/C recursion over $\mathbb{F}_3$), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey linear complexity of ($u_n \bmod 3$) is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–6. A proof for U would require a non-automaticity criterion valid for *dense*-support series — the coefficients carry no lacunary gaps — which the standard gap/Mahler methods do not provide.

8 Conclusion

The arithmetic of A396406 separates cleanly. The catalytic building blocks and the growth rate are unconditional (Pólya–Carlson; Corollary 9). Both the relaxed series V and the true series U rest on a single simple-saddle steepest-descent estimate (Lemma 4), which we now prove *self-contained* — by Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3, the two saddles $z_\pm^* = \pm iW/2$ supplying the global cancellation that no partial-summation method can — rather than by citing a general theorem. With that estimate in hand, V **is transcendental over** $\mathbb{Q}(x)$ **unconditionally**. The true series U stands on the same footing up to a single residual: through the exact identity $P_{12} = \frac{2q^3}{1-q^3}\sum_k d_k$, its one further input — the numerator amplitude at the pole, estimate $(\star)$ — is derived in elementary closed form to leading order, and the phase-matching of the travel-pole equation against the amplitude's zero gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) < 1/\sqrt{2}$: the gate holds with a fourfold margin, established to leading order and at the first 70 poles. **Conditional on the** $O(\tau)$ **residual of** $(\star)$ **— verified numerically but not proved —** U **is transcendental over** $\mathbb{Q}(x)$ with the same analytic standing as V: every coefficient derived elementarily, remainders at the standard simple-saddle level of Lemma 4. The turning-point amplitude atom on the U-side is the elementary rational bound of Lemma 11, formally verified in Lean 4. The independent mod-p route of §7 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found.

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